

CALCULUS

DERIVATIVES

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This formula is called the limit definition of the derivative.

It shows how the rate of change of a function is found when the change in x , represented by h , gets closer and closer to zero.

Differentiation is the process, while the **derivative** is the result obtained from that process

Differential Calculus and Rate of Change

Differential calculus studies derivatives and how quantities change. A derivative may be interpreted as an instantaneous rate of change or as the slope of a tangent line to a curve.

Derivative Formula

Conditions for y' or $f'(x)$:

- * No negative exponents
- * No fractional exponents

Negative Exponents

$$x^{-n} = \frac{1}{x^n}$$

$$cx^{-n} = \frac{c}{x^n}$$

Radical and Fractional Exponent

$$\sqrt[n]{x^m} = x^{\frac{m}{n}}$$

Constant Rule

A constant function does not change as x changes, so its graph is a horizontal line.

This rule is often combined with the sum, difference, and constant-multiple rules.

Given: $f(x) = c$, where c is any real number

$$f'(c) = 0$$

Linear

Given: $f(x) = cx$, where c is any real number

$$f'(x) = c$$

Power Rule

Given: $f(x) = x^n$, where n is any real number

$$f'(x^n) = \underline{nx^{n-1}}$$

Steps:

1. Change into fractional/negative exponent
2. Apply the formula
3. Simplify the exponent
4. Change into positive exponent
5. Change into radical.

Constant Multiple (Power Rule)

A constant coefficient remains as a multiplier

Given: $f(x) = cx^n$, where c and n are any real numbers

$$f'(c \cdot x^n) = c \cdot \underline{nx^{n-1}}$$

Sum/Difference Rule

$$(f + g)'(x) = f'(x) + g'(x)$$

$$(f - g)'(x) = f'(x) - g'(x)$$

Product Rule

$$d/dx (u \cdot v) = u \, dv + v \, du$$

$$f'(u \cdot v) = \underline{uv'} + vu'$$

$$(f \cdot g)'(x) = f'(x)g(x) + g'(x)f(x)$$

Quotient Rule

$$d/dx (u/v) = (vdu - u dv) / v^2$$

$$f'(u/v) = (vu' - uv') / v^2$$

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{[g(x)]^2}$$

When one differentiable function is divided by another.

Keep the order in the numerator consistent, and square the entire denominator. Simplify after applying the rule

Chain Rule

Used in finding the derivative of composition of functions, $(f \circ g)(x)$ or $f(g(x))$

Given: $y = f(g(x)) \rightarrow y' = f'(g(x)) \cdot g'(x)$

Let $y = f(u)$ and $u = g(x)$

$$dy/dx = dy/du \cdot du/dx$$

$$d/dx [f(g(x))] = f'(g(x)) \cdot g'(x)$$

Steps:

1. Identify outer function
2. Identify inner function
3. Find derivative of outer function
4. Find derivative of inner function
5. Apply the Chain Rule $(dy/du)(du/dx)$
6. Simplify
7. Substitute u

Exponential Rule

$$\text{Given: } f(x) = a^x, \text{ where } a > 0 \text{ and } a \neq 1$$

$$f'(x) = a^x \ln a$$

It means natural logarithm

$$\text{Given: } f(x) = e^x, \text{ where } a > 0 \text{ and } a \neq 1$$

$$f'(x) = e^x$$

Euler's number e is a constant, approximately equal to 2.71828

Derivatives of Logarithmic Functions

Logarithmic Rule

Given: $f(x) = \log_a x$, where $a > 0$ and $a \neq 1$

$$f'(x) = \frac{1}{x \ln a}$$

Given: $f(x) = \ln x$

$$f'(x) = \frac{1}{x}$$

Derivatives of Trigonometric Functions

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \cot x = -\csc^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \csc x = -\csc x \cot x$$

Derivative Notation

The symbol $f''(x)$ or d^2y/dx^2 represents the derivative of the first derivative.

The first derivative describes rate of change or slope, while the second derivative describes how that rate of change itself changes. It is also used in concavity and extreme tests.

Higher-Order Derivative

Repeated process of taking derivatives of derivative

n^{th} Derivative: $y^{(n)} \rightarrow f^{(n)}(x) \rightarrow d^n y/dx^n$

MAXIMA AND MINIMA

Critical Point

- ✿ Is a point on the graph of a function where the derivative is 0 or does not exist
- ✿ May be represented as turning point

Critical Number

- ✿ Given: $f'(x) = 0$
- ✿ x is a critical number

Concavity

- ✿ is the property of a function that indicates whether its graph curves upward or downward over an interval.

Concave Upward

- ❖ decreasing – increasing
- ❖ $f''(x) > 0$

Concave Downward

- ❖ increasing – decreasing
- ❖ $f''(x) < 0$

Point of Inflection

a point on the graph of a function where the concavity changes from concave up to concave down or from concave down to concave up.

Maximum - Highest point of the graph of function

Minimum - Lowest point of the graph of function

Determining the maxima and minima of a function

- ★ First derivative test
- ★ Second derivative

First Derivative Test

❖ Find the critical numbers

- ★ Get first derivative, $f'(x)$
- ★ Determine the critical numbers, $f'(x) = 0$

❖ Apply First Derivative Test

- ★ Determine the sign of $f'(x)$ on interval around each critical number
 - ★ If $f'(x)$ changes from positive to negative, the function has a **local maximum**.
 - ★ If $f'(x)$ changes from negative to positive, the function has a **local minimum**.
 - ★ If there is no sign change, there is no **local extremum**.

❖ Find the minima/maxima

- ★ Evaluate $f(x)$ at $x =$ critical number

Second Derivative Test

❖ Find the critical numbers

- ★ Get first derivative, $f'(x)$
- ★ Determine the critical numbers, $f'(x) = 0$

❖ Apply Second Derivative Test

- ★ Determine the second derivative of f , $f''(x)$
 - ★ If $f''(x_0) < 0$, the function has a **local maximum**.
 - ★ If $f''(x_0) > 0$, the function has a **local minimum**
 - ★ If $f''(x_0) = 0$, **inconclusive**

❖ Find the minima/maxima

- ★ Evaluate $f(x)$ at $x_0 =$ critical number

Handwritten notes for the First Derivative Test:

2. $f(x) = x^3 - 3x^2 - 9x + 2$

$f'(x) = 3x^2 - 6x - 9$
 $0 = 3x^2 - 6x - 9$
 $0 = 3(x^2 - 2x - 3)$
 $0 = 3(x-3)(x+1)$
 $x-3=0 \Rightarrow x=3$
 $x+1=0 \Rightarrow x=-1$
critical numbers

If Critical points when $x=3$
 $f(x) = x^3 - 3x^2 - 9x + 2$
 $= (3)^3 - 3(3)^2 - 9(3) + 2$
 $= 27 - 27 - 27 + 2$
 $f(x) = -25$ (3, -25) local pt

If Critical points when $x=-1$
 $f(x) = x^3 - 3x^2 - 9x + 2$
 $= (-1)^3 - 3(-1)^2 - 9(-1) + 2$
 $= -1 - 3 + 9 + 2$
 $= 7$ (-1, 7) critical point

Point of Inflection:
 $f''(x) = 6x - 6$
 $0 = 6x - 6$
 $-6x = -6$
 $x = 1$

1) critical number
2) critical point
3) point of inflection

